

		$d = r / \cos 40^\circ = 183.32 = 183 \text{ m}$
11	D	A: $g = -\frac{d\phi}{dr}$, while $g = 0$ at Z, it only means that potential is either a minimum or maximum at point Z. B: Since Q has a smaller magnitude of g, it has a smaller mass. C: Gravitational force is an attractive force. D: Since $g = 0$ and $F = mg$, net force at Z is zero hence the forces due to P and Q are equal in magnitude and opposite in direction

12	D	For a satellite at Earth's surface
----	---	------------------------------------